
Project 13 — Text to Speech · Presentation

Slide deck + speaker notes for the group presentation of the DL TTS project (fine-tuning **VITS**). Built for a **15–20 min talk + 5–10 min discussion**.

Source repo being presented: `~/dev/dl · github.com/Dmitri1S9/DL` Full background brief: `~/dev/dl-presentation-brief.md`

Files

File	What it is
<code>slides.pdf</code>	The deck — present from this (or <code>slides.pptx</code>).
<code>slides.pptx</code>	PowerPoint/Keynote version — edit if you want.
<code>slides.html</code>	Browser version — open + press F for fullscreen, arrows to navigate.
<code>slides.md</code>	Source (Marp Markdown). Edit this, then re-render (below).
<code>CHEATSHEET.md</code>	One-page quick-reference — print this and hold it while presenting.
<code>SPEAKER_NOTES.md</code>	Per-slide talk track with timing and speaker assignments.
<code>PRESENTATION_PLAYBOOK.md</code>	How to <i>deliver</i> it — speaker split, rehearsal, demo plan, examiner tips, rubric.
<code>DISCUSSION_QA.md</code>	~50 anticipated Q&A answers, by difficulty tier.
<code>TECHNICAL_APPENDIX.md</code>	Deep dive — VITS math/ELBO, MAS algorithm, HiFi-GAN, losses, references. For the report + hard questions.

Master background brief (the source of truth behind all of this): `~/dev/dl-presentation-brief.md`.

Suggested reading order: `CHEATSHEET` → `SPEAKER_NOTES` → `PRESENTATION_PLAYBOOK` → `DISCUSSION_QA` → `TECHNICAL_APPENDIX` (only if you want the math).

Presenting

- **Quickest:** open `slides.pdf` fullscreen, or `slides.html` in a browser (`F` = fullscreen, `←/→` = navigate).
- Read `SPEAKER_NOTES.md` first — it splits the 20 slides across the three speakers and gives cumulative timing + safety valves for over/under-running.
- For the demo (slide 17): pre-load the audio so you don't fumble in the terminal — `dl/audio/generated/test_000*.wav` , `dl/droid_test.wav` , `dl/audio/demo_*.wav` .

Re-rendering after an edit

Edit `slides.md` , then (Node + npx required):

```
cd ~/dev/dev_presentation
export CHROME_PATH="$(npx -y puppeteer browsers install chrome | awk '{print $2}')"
npx -y @marp-team/marp-cli@latest slides.md -o slides.html --html
npx -y @marp-team/marp-cli@latest slides.md -o slides.pdf --pdf --allow-local-files
npx -y @marp-team/marp-cli@latest slides.md -o slides.pptx
```

PDF/PPTX export needs a Chromium browser; the `CHROME_PATH` line downloads a self-contained one via Puppeteer (cached in `~/.cache/puppeteer` , no system install). HTML export needs no browser. Or use the **Marp for VS Code** extension for live preview + export.

Deck outline (20 slides)

1. Title · 2. The task · 3. Four decisions · 4. What is VITS · 5. Components (train vs. infer) · 6. Alignment / MAS · 7. Data story (B1 droid voice) ·
2. The core challenge (HF is inference-only) · 9. `forward_train` · 10. Five losses · 11. Discriminator (MPD+MSD) · 12. GAN training loop · 13. Evaluation ·
3. Engineering (contract-first) · 15. Status · 16. Failure modes · 17. Demo ·
4. Team · 19. Takeaways · 20. Q&A.